

MaJo e-School Bridge

Functionality Report & Actual Test Results

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1. Executive summary

Automated functionality suite executed 18 tests: 18 passed, 0 failed (100% pass). Cloud-dependent features (server login, SMS OTP, Storage bucket, FCM fan-out) are implemented in code but blocked on Firebase Blaze / deployment for live multi-device use.

2. Product surfaces

- Android app — role dashboards (teacher, parent, student, admin, driver)
- Web ERP — Flutter web admin shell (lib/web_erp)
- Backend — Firestore, Storage, Cloud Functions, Hosting

3. Functionality inventory

Implemented = code wired; Partial = gaps; Placeholder = nav without full module.

Role	Module	Description	Status	Notes / deps
Shared	Login / roles	Role-based sign-in (admin, teacher, parent, student, driver)	Implemented	Cloud login needs Blaze Functions
Shared	Forgot / change password	OTP + password reset / forced change	Implemented	Real SMS needs Blaze Phone Auth
Shared	Localization	EN / Amharic / Oromo	Implemented	None
Shared	Settings & app lock	Preferences, background lock / web idle timeout	Implemented	None
Shared	Push notifications	FCM token + local notifications	Implemented	FCM send via Functions needs Blaze
Shared	Messaging	Conversations, attachments, voice	Implemented	Cloud sync needs auth claims
Shared	Announcements	Create/view school announcements	Implemented	Cloud sync
Shared	Calendar / timetable	Events and class timetables	Implemented	Cloud sync
Teacher	Classes	My classes view	Implemented	Cloud sync
Teacher	Attendance	Take / view attendance	Implemented	Cloud sync
Teacher	Homework	Assign / manage homework	Implemented	Cloud sync
Teacher	Learning materials	Share materials	Implemented	Storage may need Blaze
Teacher	Grades	Enter grades / reports	Implemented	Cloud sync
Teacher	Gallery / QR	Gallery posts and QR tools	Implemented	Camera device
Parent	Children	Linked children overview	Implemented	Cloud sync
Parent	Fees	Fee view / payment UX (provider pending)	Partial	Online payments not live
Parent	Bus tracking	Live bus / transport status	Implemented	Cloud + location
Parent	Signup / approvals	Register + admin approval	Implemented	Register callable needs Blaze
Student	Portal	Grades, homework, materials, attendance	Implemented	Portal flags / cloud
Admin	People registries	Students / teachers / staff CRUD	Implemented	Cloud sync
Admin	Finance	Fees / finance dashboard	Implemented	Cloud sync
Admin	Transport admin	Routes, drivers, scans	Implemented	Cloud sync
Admin	Grade approvals	Workflow queue	Implemented	Cloud sync
Admin	Inventory (web)	Stock / assets module	Implemented	Storage + Blaze bucket
Admin	Web ERP shell	Sidebar ERP for web admin	Implemented	Flutter web
Admin	Institution / campus	Org management entries	Placeholder	UI placeholders
Driver	Route / passengers	Assigned route and list	Implemented	Cloud sync
Driver	QR scan / pickup	Transport scanning	Implemented	Camera
Driver	Live map share	GPS publish for parents	Implemented	Location + cloud
Platform	Owner console	Hidden PIN console	Implemented	Device-local PIN
Platform	Audit / health	Audit log and system health	Implemented	Cloud sync

4. Actual automated test report

Command: flutter test test/functionality_suite_test.dart test/database_test.dart
Exit code: 0 | Pass rate: 100%

#	Result	Test case
1	PASS	Auth & credentials debug demo admin login succeeds locally
2	PASS	Auth & credentials wrong password rejected
3	PASS	Auth & credentials role mismatch rejected
4	PASS	Auth & credentials password policy minimum is 10
5	PASS	Auth & credentials password hash verify round-trip
6	PASS	Phone utilities normalizes Ethiopian local login key
7	PASS	Phone utilities builds E.164 for Ethiopia
8	PASS	School data / access seed builds students, classes, parent links
9	PASS	School data / access parent resolver reaches child and teachers
10	PASS	School data / access teacher class access boundaries
11	PASS	School data / access driver student access boundaries
12	PASS	Platform owner PIN PIN min length is 6
13	PASS	Platform owner PIN set and verify hashed PIN
14	PASS	Web ERP surface admin nav includes core modules
15	PASS	Web ERP surface router resolves key admin pages to widgets
16	PASS	seed builds students, classes, and parent links
17	PASS	parent can reach child class teachers via resolver
18	PASS	role access rules enforce teacher and driver boundaries

5. Manual / environment checks

Check	Result	Reason
Cloud schoolLogin (phone/web)	BLOCKED	Needs Blaze + deployed Functions
Firebase Phone SMS OTP (release)	BLOCKED	Needs Blaze + Phone Auth setup
Cross-device Firestore sync	BLOCKED	Needs custom-token session
Storage upload/download	BLOCKED/PENDING	Bucket/billing + deploy rules
Maps on web	PENDING CONFIG	Set web/maps_api_key.js
Play release signing	PENDING CONFIG	android/key.properties + keystore
widget_test Login screen	FAIL (known)	Needs Firebase mock; excluded from suite

6. Role coverage (local/debug)

Capability	Admin	Teacher	Parent	Student	Driver
Local debug login	Yes	Yes	Yes	Seeded	Yes
Messages	Yes	Yes	Yes	Yes	Yes
Attendance	Yes	Yes	View	View	Transport
Grades	Approve	Enter	View	View	—
Fees	Manage	—	View*	—	—
Transport / bus	Manage	—	Track	—	Operate
Inventory (web)	Yes	—	—	—	—
Cloud (Spark today)	No	No	No	No	No

* Parent online fee providers not live.

7. Recommendations

1. Use local/debug for feature QA until Blaze.

2. Deploy Firestore + Storage rules on Spark now.
3. After Blaze: deploy Functions, bootstrap admin, re-run this report + manual cloud login.
4. Fix/mock Firebase in test/widget_test.dart for CI.
5. Add integration_test/patrol for dashboards once cloud auth is live.

Raw JSON: reports/
functionality_test_results.json
PDF: MaJo_Functionality_Test_Report.pdf